

Modernizing High-Precision 3D Scanning Software

Re-architecting a legacy Windows application into a modern, high-performance 3D scanning platform for jewelry manufacturing and CAD workflows.

Client: eLUXE3D

Project Type: Desktop Software Modernization / 3D Processing

Role: Lead Software Architect & Full-Stack Developer

Industry: Jewelry Manufacturing · 3D Scanning · CAD

Timeline: ~150–200 hours across 6 milestones

Platform: Windows Desktop

01 — Executive Summary & The Challenge

eLUXE3D develops high-precision 3D scanning technology for jewelry manufacturing. Their world-class hardware was limited by a legacy C++ architecture from ~2005. Modern scans exceeding **5 million points** caused severe memory leaks, UI freezing, application crashes, and unreliable auto-alignment.

The goal was to modernize the software architecture while preserving the raw computational power of C++. We aimed to separate the user experience from the heavy lifting, fix memory instability, and provide professional CAD inspection tools.

02 — The Solution: Split-Stack Architecture

Rather than rewriting the computational engine in JavaScript, the application was redesigned around a strict separation of responsibilities:

- **UI Layer (Electron + React + TailwindCSS):** Manages user interaction, workflow, and scanner controls.
- **Visualization Layer (Three.js + WebGL):** Real-time point-cloud rendering and 3D interaction.
- **Bridge Layer (Node.js + Node-API):** Zero-copy binary data transfer to avoid memory duplication.
- **Computational Layer (Modern C++):** Heavy lifting using Open3D and Point Cloud Library (PCL) for point-cloud processing, ICP alignment, surface reconstruction, and mesh decimation.

The result: The interface never carries the weight of the mathematics. Millions of points are processed natively without freezing the UI.

03 — Core Workflows & Hardware Control

The platform provides an end-to-end digital manufacturing workflow:

Physical Jewelry → 3D Scanner → Point Cloud → Clean & Crop → Auto Alignment → Surface Reconstruction → Mesh Optimization → CAD Export (PLY / STL / OBJ)

Key Capabilities

- **Hardware Control:** Configure optics (30mm to 80mm FOV) and mechanical sweeps visually.
- **3D Lasso Cropping:** Interactively remove unwanted mounting fixtures (e.g., blue tack) before reconstruction.
- **Automatic Scan Alignment:** Uses Iterative Closest Point (ICP) to mathematically determine optimal alignment between overlapping scans.
- **Surface Reconstruction & Decimation:** Poisson Surface Reconstruction produces high-quality continuous meshes, while Quadric Error Metrics (QEM) decimation reduces file sizes by ~90% while preserving sharp

details like prongs and gem facets.

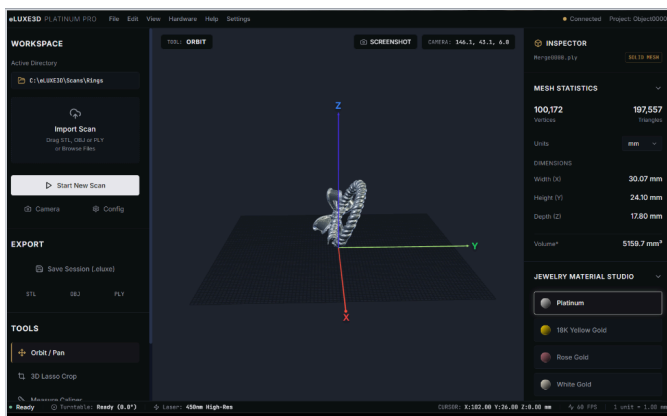
- **CAD Inspection:** Digital calipers, cross-section slicing, and realistic material rendering bridge the gap between physical jewelry and digital CAD.

04 — End-to-End Delivery

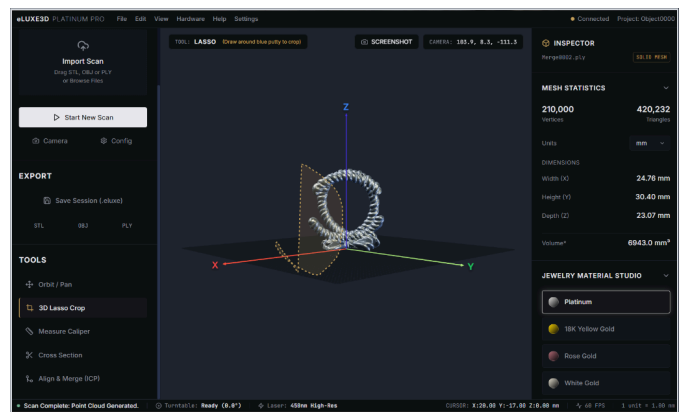
Over ~150–200 hours, I operated as the Lead Software Architect and Full-Stack Developer. I owned the complete modernization strategy, from designing the split-stack architecture and Node-API bridge to building the React/Three.js frontend and packaging the final signed Windows application with automatic OTA updates via AWS S3.

The software transformed from a legacy bottleneck into a modern foundation for high-precision 3D scanning.

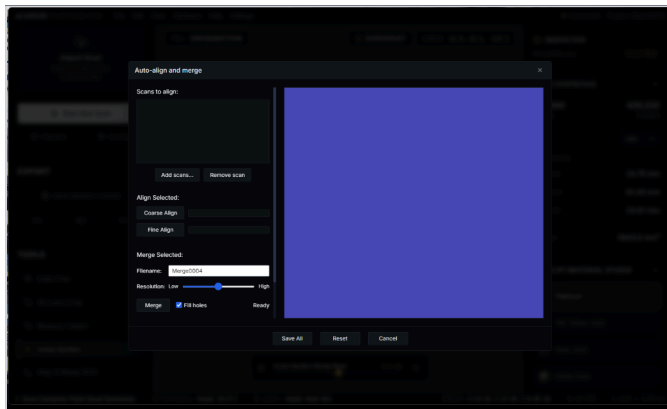
05 — Visual Product Showcase



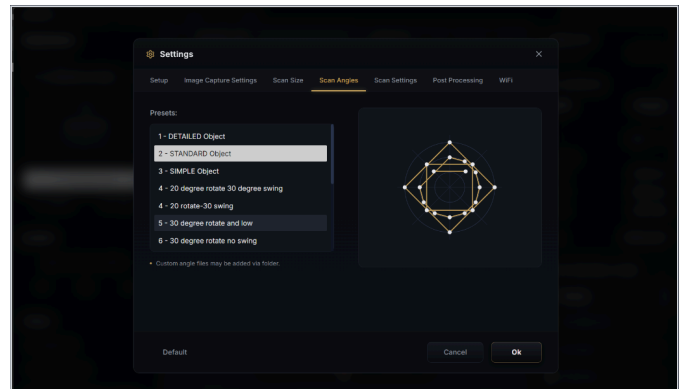
Main scanning workspace with real-time mesh stats and jewelry material rendering.



3D lasso selection around unwanted fixtures before deletion.



Automated ICP-based alignment of multiple scans into a unified model.



Polar radar / scan-angle visualization for custom mechanical sweep paths.

Confidentiality note

Certain proprietary mathematical configurations, hardware-driver implementations, and client-specific technical details are excluded from this case study in accordance with confidentiality requirements.